AN LCI MAP INTO A REGULAR S-SCHEME FORCES THE MAP AND THE TARGET'S STRUCTURE MORPHISM TO BE KOSZUL AT THE IMAGE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18420-18434 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18436-18440, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Specialization-of-named-theorem template (one displayed deduction from three inputs); accepted.

Allowed standard prerequisites (editorial).

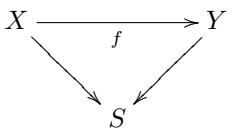
- Explicit prerequisite boundary. Chapter lemma lci-avramov (the Avramov correspondence, card L16065); chapter lemma regular-target-perfect; Morphisms, lemma permanence-finite-type.

Source reference. Original lemma 09RL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18357–18371. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17802–17806. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18440–18440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume S is locally Noetherian, $Y \to S$ is locally of finite type, Y is regular, and $X \to S$ is a local complete

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intersection morphism. Then $f : X \to Y$ is a local complete intersection morphism and $Y \to S$ is Koszul at $f(x)$ for all $x \in X$.

Proof. This is a special case of Lemma 09RL in view of Lemma 068B (and Morphisms, Lemma 01T8). □